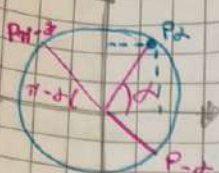
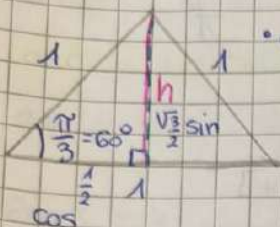


17/09/25



$$P_{\alpha} = (\cos \alpha, \sin \alpha)$$

$$\cos \frac{\pi}{4} = \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$$



• l'altezza di un triangolo non cade sempre sul punto medio

$$1 = \sqrt{\frac{1}{4} + x^2}$$

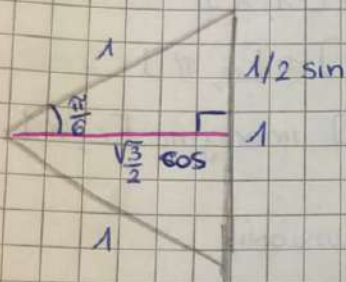
$$1 = \frac{1}{4} + x^2$$

$$x^2 = 1 - \frac{1}{4} = \frac{3}{4} \rightarrow x = \frac{\sqrt{3}}{2}$$

$$\cos \frac{\pi}{3} = \frac{1}{2}$$

$$\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

• Le distanze sono sempre positive



$$\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$$

$$\sin \frac{\pi}{6} = \frac{1}{2}$$

$$\cos^2 \alpha + \sin^2 \alpha = 1$$

$$\cos(-\alpha) = \cos \alpha$$

$$|\cos \alpha| \leq 1 \text{ e } |\sin \alpha| \leq 1$$

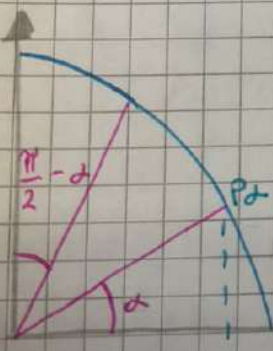
$$\sin(-\alpha) = -\sin \alpha$$

$$\cos(\alpha + 2\pi k) = \cos \alpha$$

$$\cos(\pi - \alpha) = -\cos \alpha$$

$$\sin(\alpha + 2\pi k) = \sin \alpha$$

$$\sin(\pi - \alpha) = \sin \alpha$$



$$\cos\left(\frac{5}{6}\pi\right) = \cos\left(\pi - \frac{\pi}{6}\right) = -\cos\left(\frac{\pi}{6}\right) = -\frac{\sqrt{3}}{2}$$

$$\sin\left(\frac{5}{6}\pi\right) = \sin\left(\pi - \frac{\pi}{6}\right) = \sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

$$\sin\left(\frac{3}{4}\pi\right) = \frac{\sqrt{2}}{2}$$

$$\cos\left(\frac{3}{8}\pi\right) = \cos\left(\frac{\pi}{2} - \frac{\pi}{8}\right) = \sin\left(\frac{\pi}{8}\right)$$

DUPLICAZIONE

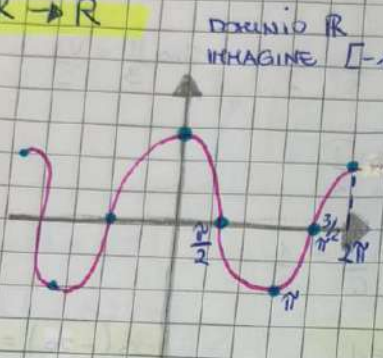
$$\begin{aligned} \cos(2\alpha) &= \cos^2 \alpha - \sin^2 \alpha \\ &= 1 - 2\sin^2 \alpha \\ &= 2\cos^2 \alpha - 1 \end{aligned}$$

$$\sin(2\alpha) = 2\cos \alpha \sin \alpha$$

$$(\cos \alpha + \sin \alpha)^2 = 1 + \sin(2\alpha)$$

α	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\cos \alpha$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\sin \alpha$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\frac{\sin \alpha}{\cos \alpha} = \tan \alpha$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	$\nexists$

$\cos x: \mathbb{R} \rightarrow \mathbb{R}$



DOMINIO $\mathbb{R}$
IMMAGINE $[-1; 1]$ $\cos x: \mathbb{R} \rightarrow [-1; 1]$

strett. cresc. in $[\pi; 2\pi]$

strett. decresc. in $[0; \pi]$

$\cos x: [0; \pi] \rightarrow [-1; 1]$

$\arccos x: [-1; 1] \rightarrow [0; \pi]$

dominio $[-1; 1]$ immagine $[0; \pi]$

- $\arccos x$ e' l'angolo il cui coseno e' x

$$\arccos\left(\frac{1}{2}\right) = \frac{\pi}{3}$$

$\arccos x > 5 \rightarrow$ NON HA SOLUZIONI

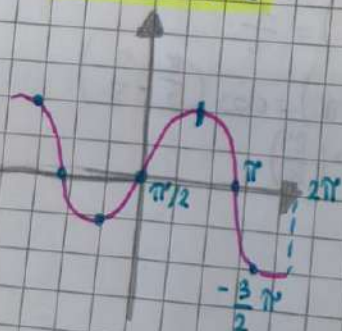
$\arccos(400) \rightarrow$ NON ESISTE

$\arccos(-10) \rightarrow$ NON ESISTE

$\arccos(75) \rightarrow$ NON ESISTE

$\arccos(\pi) \rightarrow$ NON ESISTE

$\sin x: \mathbb{R} \rightarrow \mathbb{R}$



DOMINIO $\mathbb{R}$

IMMAGINE $[-1; 1]$

$\sin x: \mathbb{R} \rightarrow [-1; 1]$

periodica di periodo 2π

$\arcsin x: [-1; 1] \rightarrow \left[-\frac{\pi}{2}; \frac{\pi}{2}\right]$

$$\arcsin\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3}$$

$\arcsin(100)$

$\arcsin(10)$

$\arcsin(3\pi)$

NON ESISTE

$$E_{\text{og}}(4 \cos x \sin x) > 0$$

$$\log_2(2 \sin(2x)) > 0 \iff \begin{cases} 2 \sin(2x) > 1 \\ \sin(2x) > 1/2 \end{cases}$$

$$\frac{\pi}{6} + 2k\pi < 2x \leq \frac{\pi}{2} + 2k\pi \rightarrow \frac{\pi}{12} + k\pi < x \leq \frac{\pi}{4} + k\pi$$

$$\frac{\sin x}{\sin(2x)} > 1$$

$\sin[0, \pi]$

$$\begin{aligned} x &\neq 0 \\ x &\neq \pi/2 \\ x &\neq \pi \end{aligned}$$

$$\frac{\sin x}{2 \cos x \sin x} > 1$$

2. $\frac{1}{2 \cos x} > 1 \rightarrow \frac{1}{2 \cos x} - 1 > 0 \rightarrow \frac{1 - 2 \cos x}{2 \cos x} > 0$

$\frac{2\cos x}{x}$

0 ++++++ $\frac{\pi}{2}$ ----- π D

$$\frac{\pi}{3} < x < \frac{\pi}{2}$$

$$\bullet 3(\sin x + 1) < \cos^2 x \rightarrow 3\sin x + 3 < 1 - \sin^2 x \rightarrow \sin^2 x + 3\sin x + 2 < 0$$

$$t = \sin x$$

$$t^2 + 3t + 2 < 0$$

$$\frac{-3 \pm \sqrt{9-8}}{2} \rightarrow \begin{matrix} -1 \\ -2 \end{matrix}$$

$$-2 < t < -1$$

$$-2 < \sin x < -1$$

$$\bullet \begin{cases} x(y^2 - x^2) = 0 \\ x^2 + y^2 = 2 \end{cases} \iff \begin{cases} x = 0 \\ x^2 + y^2 = 2 \end{cases} \cup \begin{cases} y^2 - x^2 = 0 \\ x^2 + y^2 = 2 \end{cases}$$

$$\begin{pmatrix} 0, \sqrt{2} \\ 0, -\sqrt{2} \end{pmatrix}$$

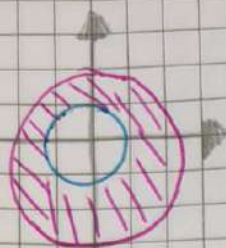
$$\begin{pmatrix} 0, \sqrt{2} \\ 0, -\sqrt{2} \end{pmatrix} \quad \begin{cases} x=0 \\ y^2=2 \end{cases}$$

$$\begin{cases} x^2 = 1 \\ y^2 = 1 \end{cases}$$

$$\begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}$$

$$V \begin{cases} y^2 - x^2 = 0 \\ x^2 + y^2 = 2 \end{cases}$$

$$\vee \begin{cases} x^2 = y^2 \\ 2x^2 = 2 \end{cases} \rightarrow \begin{cases} y^2 = 1 \\ x^2 = 1 \end{cases}$$



$$x^2 + y^2 = 1$$

$$x^3 + 4y^2 + 5x$$

$$x^2 + y^2 = 4$$

$$1 \leq x^2 + y^2 \leq 4$$

$$\begin{cases} \log(x^2) - \log y = 0 \\ x(\sqrt{y} - x) = 0 \end{cases}$$

$$\boxed{\begin{matrix} x \neq 0 \\ y > 0 \end{matrix}}$$

C.E.

$$\begin{cases} \log\left(\frac{x^2}{y}\right) = 0 \\ x(\sqrt{y} - x) = 0 \end{cases} \iff \begin{cases} \frac{x^2}{y} = 1 \rightarrow x^2 = y \\ x(\sqrt{y} - x) = 0 \\ \sqrt{y} = x \end{cases}$$

$$\{(x; x^2) \in \mathbb{R}^2 : x \in \mathbb{R}\} \setminus \{0\}$$

$$\{(x; y) \in \mathbb{R}^2 : y = x^2, x \neq 0\}$$

$$\begin{cases} \log_3(3x) - \log_3 x = y \\ e^{x+y} = +\frac{1}{2} \end{cases}$$

$$\boxed{x > 0}$$

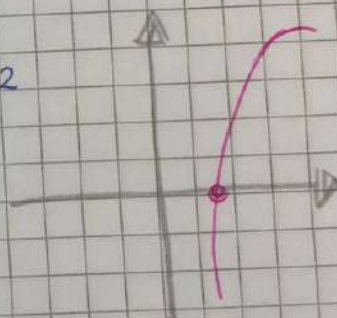
$$\log_3\left(\frac{3x}{x}\right) = y \rightarrow y = 2$$

$$x = -\frac{1}{2} + \log \frac{1}{2}$$

sol

$$e^{x+2} = \frac{1}{2}$$

$$x+2 = \log \frac{1}{2}$$



$$\frac{\sin x}{\cos x} = \tan x : D \rightarrow \mathbb{R}$$

$$D = \left\{ x \in \mathbb{R} \mid x \neq \frac{\pi}{2} + k\pi, k \in \mathbb{Z} \right\}$$

IMMAGINE: $\mathbb{R}$

periodica di periodo π

$$\arctan x : \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

• $\arctan x$ e' l'angolo cui Tangente e' x

$$\arctan(-1) = -\frac{\pi}{4}$$

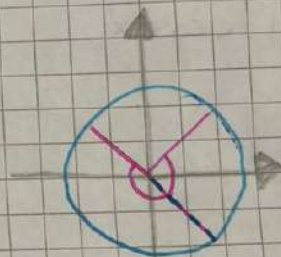
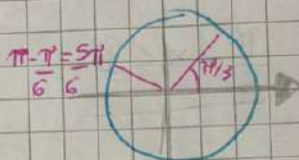
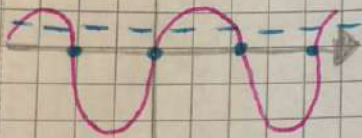
$$\begin{aligned} \log(\cos x) > 0 \\ e \downarrow e \\ \cos x > 1 \quad \text{non sol} \end{aligned} \quad \left\{ \begin{array}{l} \cos x > 0 \\ \cos x > 1 \end{array} \right.$$

$$\bullet \sin x > 2 \rightarrow \nexists x$$

$$\bullet \sin x > -\pi \rightarrow \forall x \in \mathbb{R}$$

$$\bullet \sin x > 1/2 \rightarrow \frac{\pi}{6} + 2k\pi < x < \frac{5\pi}{6} + 2k\pi, k \in \mathbb{Z}$$

$$\bullet \sin x < \frac{\sqrt{3}}{2}$$



$$\bullet \cos(2x) - \sin(x) = 0$$

$$\begin{aligned} 1 - 2\sin^2 x \\ - 2\sin^2 x - \sin x + 1 = 0 \\ - 2t^2 - t + 1 = 0 \\ \frac{1 \pm \sqrt{1+8}}{-4} \end{aligned}$$

$$\begin{aligned} t &= -1 \\ t &= 1/2 \end{aligned}$$

$$t = \sin x$$

$$\sin x = -1 \rightarrow x = \frac{3\pi}{2} + 2k\pi$$

$$\sin x = 1/2 \rightarrow x = \frac{\pi}{6} + 2k\pi$$

$$\begin{aligned} 2k\pi - \frac{\pi}{3} < x < \frac{\pi}{3} \\ 2k\pi + \frac{2\pi}{3} < x < \pi + 2k\pi \end{aligned}$$

$$\begin{aligned} |\sin x|^2 &= |\sin x| \cdot |\sin x| \\ &= |\sin^2 x| = \sin^2 x \end{aligned}$$

$$\bullet \sqrt{1 - \cos x} \leq |\sin x|$$

$$\begin{aligned} 1 - \cos x \geq 0 &\rightarrow 1 - \cos x \leq \sin^2 x \\ \cos x &\leq 1 \end{aligned}$$

$$\begin{aligned} 1 - \cos x &\leq 1 - \cos^2 x \\ \cos x &\geq \cos^2 x \end{aligned}$$

$$\bullet \cos^2 x - \cos x \leq 0$$

$$t = 0 \quad 0 \leq \cos x \leq 1 \quad t = \cos x$$

$$t = 1 \quad t^2 - t \leq 0$$

